

CRF Fluctuation Law v3 (Final)

$$Q_{\text{true}} = \sigma_N / \sqrt{\bar{N}} \approx 1 / \sqrt{\kappa n T_{\text{avg}}}$$

Unit Correction, 5-Seed Confirmation, and Correlation Structure

Supersedes v1 and v2. Canonical definition fixed.

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Abstract

We establish the CRF Fluctuation Law in its correct form and identify the correlation structure responsible for the residual 5.2% gap.

Unit correction. v1/v2 computed $Q_A = \sqrt{\kappa} Q_{\text{true}}$; the prediction also lacked $\sqrt{\kappa}$. Both errors cancelled, producing a spurious 1.0% match.

Corrected law (Hypothesis). $Q_{\text{true}} \equiv \sigma_N / \sqrt{\bar{N}} \approx 1 / \sqrt{\kappa n T_{\text{avg}}} = 0.15338$.

Confirmation (5 seeds). $Q_{\text{true}} = 0.16129 \pm 0.00227$ (CV 1.4%, gap 5.2%).

Correlation structure. Global $\text{Cov}_{\text{all}} = -6.6 \times 10^{-4}$ (0.20% of Var, too small to explain gap). Neighbour $\text{Cov}_{\text{nb}} = +0.012$ (positive – nearby nodes fire together). The implied correction requires $|\text{Cov}_{\text{implied}}| \approx 0.025$, which is $40\times$ larger than Cov_{all} . The correlation is spatially structured, not mean-field.

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1 Canonical Definition

Canonical Quantities

N_i = total R-events of node i ; $\bar{N} = \text{mean}(N_i)$; $\Omega_i = \kappa N_i$.

$$Q_{\text{true}} \equiv \frac{\sigma_N}{\sqrt{\bar{N}}} = \frac{\text{std}(N_i)}{\sqrt{\text{mean}(N_i)}}$$

Invalid forms: $\sigma_\Omega/\text{mean}(\Omega)$ or cumulative rate without proper normalisation.

2 Unit Error in v1/v2

The Error: Hidden $\sqrt{\kappa}$ Factor

v1/v2 used $\bar{\Omega}_A = \text{cumR}[s]/N = \text{mean}(N_{\text{eff}})$ (mean event count, not mean mass). This gave $Q_A = \sqrt{\kappa} Q_{\text{true}}$. The prediction used $1/\sqrt{nT}$ instead of $1/\sqrt{\kappa nT}$. Both off by $\sqrt{\kappa} = 0.387$, cancelling to give spurious 1.0%.

Quantity	v1 value	True value	Ratio
Q measured (5-seed)	0.05880	0.16129	2.74
Q prediction	0.05941	0.15338	2.58
Expected $1/\sqrt{\kappa}$	—	2.582	—

3 The Corrected Law

Fluctuation Law v3 (Hypothesis)

$$Q_{\text{true}} \equiv \frac{\text{std}(N_{\text{eff},i})}{\sqrt{\text{mean}(N_{\text{eff},i})}} \approx \frac{1}{\sqrt{\kappa n T_{\text{avg}}}} = \frac{1}{\sqrt{0.15 \times 8 \times 35.42}} = 0.15338$$

Physical meaning: $Q = 1$ for independent Poisson; CRF gives $Q \approx 0.16 \ll 1$. Suppression factor $1/Q^2 \approx \kappa n T_{\text{avg}} \approx 42$. Each of $n = 8$ SO(3) modes must accumulate T_{avg} sweeps of KL drift; mass coupling κ scales the energy per event. Status: **Hypothesis**. Step 2 (independence of inter-event periods) is unproved.

4 5-Seed Confirmation

$Q_{\text{true}} = 0.16129 \pm 0.00227$, 5 Seeds

CRF Hybrid V20.5, $N = 500$, $\kappa = 0.15$, 800 sweeps. Analysis: last 40%.

Seed	Q_{true} mean	Std	Late slope
42	0.15938	0.00479	-2.1×10^{-4}
123	0.15965	0.00343	-1.7×10^{-4}
777	0.16089	0.00478	-0.2×10^{-4}
2024	0.16565	0.00416	-2.9×10^{-4}
99	0.16089	0.00369	$+0.1 \times 10^{-4}$
Mean	0.16129	0.00227	all $< 3 \times 10^{-4}$

Prediction $1/\sqrt{\kappa n T} = 0.15338$. Gap: 5.2%. CV across seeds: 1.4%.

5 Correlation Structure

5.1 Direct measurement

Quantity	Value	Interpretation
Cov_{all} (all pairs)	-6.6×10^{-4}	anti-correlation exists
$\text{Cov}_{\text{all}}/\text{Var}$	0.20%	too small to explain gap
Cov_{nb} (KL-neighbours)	+0.012	nearby nodes fire together
$\text{Cov}_{\text{implied}}$ from gap	-0.025	40× larger than Cov_{all}

5.2 Interpretation

Structured Correlation Field, Not Mean-Field

Mean-field $\text{Cov}_{\text{all}} = -6.6 \times 10^{-4}$ is 40× too small to explain the 5.2% gap. Single-node drift-diffusion also predicts the wrong sign for $Q(\kappa)$.

$\text{Cov}_{\text{nb}} = +0.012 > 0$: nearby nodes (in KL-space) fire together. Global mass conservation forces compensating anti-correlation at long range.

The correlation is spatially heterogeneous, not mean-field:

$$C(r) = \text{Cov}(N_i, N_j \mid d_{ij} = r) > 0 \text{ (short range), } < 0 \text{ (long range).}$$

Step 2 proof requires understanding this spatial structure, not its global mean.

6 Status Table

Result	Status	Note
$Q_{\text{true}} = 0.16129 \pm 0.00227$ (5 seeds)	Confirmed	CV 1.4%
Q_{true} stationary	Confirmed	all slopes $< 3 \times 10^{-4}$
Suppression $1/Q^2 \approx \kappa n T \approx 42$	Confirmed	numerical
Unit error in v1/v2	Corrected	$Q_A = \sqrt{\kappa} Q_{\text{true}}$
v1 1.0% gap was spurious	Confirmed	errors cancelled
$Q_{\text{true}} = 1/\sqrt{\kappa n T}$ (5.2%)	Hypothesis	Step 2 unproved
$\text{Cov}_{\text{all}} < 0$	Confirmed	0.20% of Var
$\text{Cov}_{\text{nb}} > 0$ (local clustering)	Confirmed	+0.012
Mean-field Cov explains gap	Falsified	$40\times$ too small
Correlation spatially structured	Hypothesis	local+/global–
$Q(\kappa) \propto \kappa^{0.274}$	Hypothesis	$R^2 = 0.95$, not derived
$C(r)$ spatial covariance	Open	next experiment
Step 2 proof via $C(r)$	Open	Priority 1

7 Open Questions

Next Targets

1. **Spatial $C(r)$ (Priority 1).** Measure $C(r) = \text{Cov}(N_i, N_j \mid d_{ij} = r)$ vs KL-graph distance. Prediction: $C(1) > 0$ (confirmed), $C(r) < 0$ at large r . If $C(r)$ decays exponentially with scale ξ_c : derive ξ_c from the δ -chain to close Step 2.
2. **Cluster variance (Priority 2).** Group into KL clusters; measure $\text{Var}(N_{\text{cluster}})$. If suppression stronger within clusters: spatial structure confirmed.
3. **Derive $\kappa^{0.274}$ (Priority 2).** Path through $\text{Var}(\lambda_i)$ and $\theta_{\text{eff}} = K^* \sqrt{1 + \kappa N_{\text{eff}}}$.

The error was found. The law survived it.

$Q_A = \sqrt{\kappa} Q_{\text{true}}$: two equal mistakes, one apparent precision.

The corrected gap is 5.2%, not 1.0%.

That is a more honest number — and a more useful one.

Global anti-correlation exists but is $40\times$ too small.

*Nearby nodes fire together. Far nodes compensate.
The correlation is not a number. It is a field.
The δ -chain has not finished speaking.*